

随机过程作业二参考答案

习题三 3, 5, 8

3、设 $X_1, X_2, \dots$ 独立同分布, $P(X_i = 1) = p = 1 - P(X_i = 0), 0 < p < 1$. 对 $n \geq 1$, 令

$$L_n = \begin{cases} 0, & X_n = 0 \\ \max\{1 \leq k \leq n : X_{n+1-1} X_{n+1-2} \cdots X_{n+1-k} = 1\}, & X_n = 1 \end{cases}$$

为第 n 次出现的 1 的游程长度。例如, $(X_1, X_2, X_3, X_4, X_5) = (1, 0, 1, 1, 1)$, 那么对应的 $(L_1, L_2, L_3, L_4, L_5) = (1, 0, 1, 2, 3)$ 则 $\{L(n)\}$ 是一时齐马尔可夫链, 写出它的状态空间和一步转移概率。

解:

$$L_{n+1} = \begin{cases} 0, & X_{n+1} = 0 \\ L_n + 1, & X_{n+1} = 1 \end{cases} \quad p_{i(i+1)} = p \quad p_{i0} = 1 - p, \forall i \in I$$

5、独立重复掷骰子, 令 X_n 表示第 n 次得到的点数, 令 $Y_n = \max\{X_{n+1}, X_{n+2}\}, Z_n = X_{n+1} + X_{n+2}, \forall n \geq 0$

(1) 计算 $P(Y_2 = 1 | Y_0 = 1, Y_1 = 6), P(Y_2 = 1 | Y_1 = 6)$;

(2) 计算 $P(Z_2 = 12 | Z_0 = 2, Z_1 = 7), P(Z_2 = 12 | Z_1 = 7)$;

(3) 判断 $\{Y_n\}, \{Z_n\}$ 是否具有马尔可夫性? 说明理由。

解:

(1)

$$X_1 = X_2 = 1, X_3 = 6, Y_2 = 6$$

$$P(Y_2 = 1 | Y_0 = 1, Y_1 = 6) = 0$$

$$X_3 = X_4 = 1, X_2 = 6, \max\{X_2, X - 3\} = 6$$

$$P(Y_2 = 1 | Y_1 = 6) = \frac{1}{6} \times \frac{1}{2 \times 6 - 1} = \frac{1}{66}$$

(2)

$$P(Z_2 = 12 | Z_0 = 2, Z_1 = 7) = \frac{1}{6}$$

$$X_3 = X_4 = 6, X_2 = 1, X_2 + X_3 = 7$$

$$P(Z_2 = 12 | Z_1 = 7) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

(3)

$$P(Y_2 = 1 | Y_0 = 1, Y_1 = 6) \neq P(Y_2 = 1 | Y_1 = 6)$$

$$P(Z_2 = 12 | Z_0 = 2, Z_1 = 7) \neq P(Z_2 = 12 | Z_1 = 7)$$

$\{Y_n\}, \{Z_n\}$ 未来受过去影响, 不具备马尔可夫性。

8、设 $\{X_n\}$ 是一时齐马尔可夫链, 状态空间为 $\{0, 1, 2\}$, 一步转移矩阵为

$$P = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ 0 & 1 & 0 \end{bmatrix}$$

设 $P(X_0=0)=P(X_0=1)=P(X_0=2)=\frac{1}{3}$.

- (1) 计算 $P(X_2=0|X_0=0), P(X_0=0|X_2=0)$;
 (2) 计算 $P(X_1=0), P(X_1=0, X_3=0, X_4=1, X_6=1)$;
 (3) 计算 $f_{11}^{(n)}, f_{11}, \mu_1$

解:

(1)

$$P(X_0=0)=\frac{1}{3}$$

$$P(X_0=0, X_2=0)=\frac{1}{3} \times \left(\frac{1}{3} \times \frac{1}{3} + \frac{2}{3} \times \frac{2}{3} + 0 \right) = \frac{5}{27}$$

$$P(X_2=0|X_0=0)=\frac{P(X_0=0, X_2=0)}{P(X_0=0)}=\frac{5}{9}$$

计算 P 的二次幂:

$$P^2 = \begin{bmatrix} \frac{5}{9} & \frac{2}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{7}{9} & 0 \\ \frac{2}{9} & 0 & \frac{1}{3} \end{bmatrix}$$

$$P(X_2=0)=P(X_0=0)p_{00}^2+P(X_0=1)p_{10}^2+P(X_0=2)p_{20}^2=\frac{1}{3} \times \frac{5}{9} + \frac{1}{3} \times \frac{2}{9} + \frac{1}{3} \times \frac{2}{9} = \frac{13}{27}$$

$$P(X_0=0|X_2=0)=\frac{P(X_2=0, X_0=0)}{P(X_2=0)}=\frac{5}{13}$$

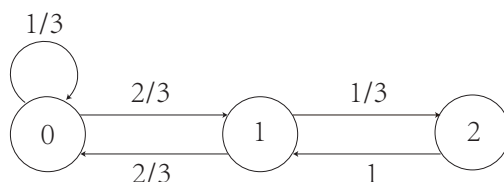
(2)

$$P(X_1=0)=\frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{3} = \frac{1}{3}$$

$$P(X_1=0, X_3=0, X_4=1, X_6=1)=\frac{1}{3} \times \frac{5}{9} \times \frac{2}{3} \times \frac{7}{9} = \frac{70}{729}$$

(3)

画出马尔可夫链的状态转移图:



$$f_{11}^{(1)}=0$$

$$f_{11}^{(2)}=\frac{2}{3} \times \frac{2}{3} + \frac{1}{3} \times 1 = \frac{7}{9}$$

$$f_{11}^{(n)}=\frac{2}{3} \times \frac{2}{3} \times \left(\frac{1}{3}\right)^{n-2} = 4\left(\frac{1}{3}\right)^n, n \geq 3$$

$$f_{11}=\frac{7}{9} + \sum_{n=3}^{+\infty} 4\left(\frac{1}{3}\right)^n = \frac{7}{9} + \frac{4/27}{1-1/3} = 1$$

$$\mu_1=\frac{14}{9} + \sum_{n=3}^{+\infty} 4n\left(\frac{1}{3}\right)^n = \frac{14}{9} + \frac{7}{9} = \frac{7}{3}$$